Experiment 1 ? Program 7

DATA SCIENCE - 22DS102006

Data Aggregation

Create a tabular dataset as a DataFrame and split data into groups using groupby method including single key and multiple key values. Select group by considering single and multiple columns.

Compute summary statistics such as sum, mean and standard deviation for the grouped data using aggregate method.

Use groupby function to split data into groups based on one column, miultiple columns, compute summary statistics and perform exploratory data analysis. Consider any online dataset for processing.

Create a tabular dataset as a DataFrame and split data into groups using groupby method including single key and multiple key values. Select group by considering single and multiple columns.

```
import pandas as pd

# Create a tabular dataset

data = {

'Name': ['Ravi', 'Sita', 'Arun', 'Priya', 'Kiran', 'Anil', 'Lakshmi', 'Rahul'],

'Department': ['CSE', 'CSE', 'ECE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],

'Year': [2, 2, 3, 3, 2, 3, 3, 2],

'Marks': [85, 90, 78, 88, 75, 82, 95, 70]

}

# Create DataFrame

df = pd.DataFrame(data)

print("Original DataFrame:")

print(df)

*/ # 1. Group by a SINGLE KEY */

print("\n1. Group by Department:")

group1 = df.groupby('Department')

for name, group in group1:

print("\nDepartment:", name)
```

```
print(group)
# Calculate average marks for each department
print("\nAverage Marks by Department:")
print(df.groupby('Department')['Marks'].mean())
*/ # 2. Group by MULTIPLE KEYS */
print("\n2. Group by Department and Year:")
group2 = df.groupby(['Department', 'Year'])
for name, group in group2:
print("\nGroup:", name)
print(group)
# Calculate average marks for each Department and Year
print("\nAverage Marks by Department and Year:")
print(df.groupby(['Department', 'Year'])['Marks'].mean())
```

Output:

Original DataFrame:

Name Department Year Marks

0 Ravi CSE 2 85

1 Sita CSE 2 90

2 Arun ECE 3 78

3 Priya ECE 3 88

4 Kiran CSE 2 75

5 Anil ECE 3 82

6 Lakshmi CSE 3 95

7 Rahul ECE 2 70

1. Group by Department:

Department: CSE

Name Department Year Marks

0 Ravi CSE 2 85

1 Sita CSE 2 90

4 Kiran CSE 2 75

6 Lakshmi CSE 3 95

Department: ECE

Name Department Year Marks

2 Arun ECE 3 78

3 Priya ECE 3 88

5 Anil ECE 3 82

7 Rahul ECE 2 70

Average Marks by Department:

Department

CSE 86.25

ECE 79.50

Name: Marks, dtype: float64

2. Group by Department and Year:

Group: ('CSE', 2)

Name Department Year Marks

0 Ravi CSE 2 85

1 Sita CSE 2 90

4 Kiran CSE 2 75

Group: ('CSE', 3)

Name Department Year Marks

6 Lakshmi CSE 3 95

Group: ('ECE', 2)

Name Department Year Marks

7 Rahul ECE 2 70

Group: ('ECE', 3)

Name Department Year Marks

2 Arun ECE 3 78

3 Priya ECE 3 88

5 Anil ECE 3 82

Average Marks by Department and Year:

Department Year

CSE 2 83.333333

3 95.000000

ECE 2 70.000000

3 82.666667

Name: Marks, dtype: float64

Compute summary statistics such as sum, mean and standard deviation for the grouped data using aggregate method.

```
import pandas as pd

# Create a DataFrame
data = {
 'Department': ['CSE', 'CSE', 'CSE', 'ECE', 'ECE', 'ECE',
 'EEE', 'EEE', 'EEE'],
 'Marks': [80, 75, 85, 70, 65, 75, 90, 85, 95],
 'Attendance': [90, 85, 95, 80, 75, 85, 95, 90, 100]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Group data by Department and calculate summary statistics
result = df.groupby('Department').aggregate({
 'Marks': ['sum', 'mean', 'std'],
 'Attendance': ['sum', 'mean', 'std']
})

print("\nSummary Statistics:")
print(result)
```

Output:

Original DataFrame:

Department Marks Attendance

0 CSE 80 90

1 CSE 75 85

2 CSE 85 95

3 ECE 70 80

4 ECE 65 75

5 ECE 75 85

6 EEE 90 95

7 EEE 85 90

8 EEE 95 100

Summary Statistics:

Marks Attendance

sum mean std sum mean std

Department

CSE 240 80.0 5.0 270 90.000000 5.000000

ECE 210 70.0 5.0 240 80.000000 5.000000

EEE 270 90.0 5.0 285 95.000000 5.000000

Use groupby function to split data into groups based on one column, miultiple columns, compute summary statistics and perform exploratory data analysis. Consider any online dataset for processing.

```
import pandas as pd
# Read the online Iris dataset
url="https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv"
df = pd.read_csv(url)

# Display first 5 records
print("First 5 Records:")
```

```
print(df.head())

# Display dataset information
print("\nDataset Information:")
print(df.info())

# Display statistical summary
print("\nStatistical Summary:")
print(df.describe())

# 1. GroupBy using a single column
print("\n1. GroupBy using a Single Column:")
group_single = df.groupby('species')['sepal_length'].agg(
 ['count', 'sum', 'mean', 'std', 'min', 'max']
)
print(group_single)

# 2. GroupBy using multiple columns

print("\n2. GroupBy using Multiple Columns:")
group_multiple = df.groupby(
 ['species', 'sepal_width']
)['sepal_length'].mean()
print(group_multiple.head(15))

# 3. Summary statistics for each species

print("\n3. Summary Statistics:")
summary = df.groupby('species').agg({
 'sepal_length': ['mean', 'std', 'min', 'max'],
 'sepal_width': ['mean', 'std', 'min', 'max'],
 'petal_length': ['mean', 'std', 'min', 'max'],
 'petal_width': ['mean', 'std', 'min', 'max']
})
```

```
print(summary)

# 4. Exploratory Data Analysis

print("\n4. Number of Records in Each Species:")
print(df['species'].value_counts())

print("\nAverage Petal Length by Species:")
print(df.groupby('species')['petal_length'].mean())

print("\nAverage Petal Width by Species:")
print(df.groupby('species')['petal_width'].mean())

print("\nMaximum Sepal Length by Species:")
print(df.groupby('species')['sepal_length'].max())
```

Output:

Sample Output

First 5 Records:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

GroupBy Using a Single Column

	count	sum	mean	std	min	max
species						

setosa 50 250.3 5.006 0.352490 4.3 5.8

versicolor 50 296.8 5.936 0.516171 4.9 7.0

virginica 50 329.4 6.588 0.635880 4.9 7.9

GroupBy Using Multiple Columns

species sepal_width

setosa 2.3 5.00

2.4 5.05

2.5 5.12

2.6 5.00

2.7 5.10

2.8 5.20

2.9 5.30

3.0 4.90

3.1 4.90

3.2 4.70

...

Summary Statistics

sepal_length sepal_width

mean std min max mean std

species

setosa 5.006 0.352490 4.3 5.8 3.428 0.379

versicolor 5.936 0.516171 4.9 7.0 2.770 0.314

virginica 6.588 0.635880 4.9 7.9 2.974 0.322

Exploratory Data Analysis

Number of Records in Each Species:

setosa 50

versicolor 50

virginica 50

Name: count, dtype: int64

Average Petal Length by Species:

setosa 1.462

versicolor 4.260

virginica 5.552

Name: petal_length, dtype: float64

Average Petal Width by Species:

setosa 0.246

versicolor 1.326

virginica 2.026

Name: petal_width, dtype: float64

Maximum Sepal Length by Species:

setosa 5.8

versicolor 7.0

virginica 7.9

Name: sepal_length, dtype: float64

7. Data Aggregation

a. GroupBy and Aggregation

```
df = pd.DataFrame({'Category': ['A', 'A', 'B', 'B'],
'Data': [10, 20, 30, 40]})
grouped = df.groupby('Category')
print(grouped['Data'].sum())
```

b. Multiple Aggregates

```
print(grouped['Data'].agg(['mean', 'std']))
```